Reproducibility Budgets for Open-Weight LLM Evaluation: A Generalizability Theory Analysis

Anonymous ACL submission

Abstract

Large Language Model (LLM) evaluation results vary across reproductions, but variance sources interact, and their relative importance shifts with the benchmark format. No unified factorial framework exists to decompose these sources simultaneously across benchmarks. We apply Generalizability theory (G-theory) in a crossed design with up to seven facets across eight open-weight 7–9B parameter models and five benchmarks, totaling 2.3 million evaluation records. Our results suggest a format-associated variance pattern: on multiple-choice benchmarks, item-related variance—item difficulty plus model×item differential item functioning (DIF)—concentrates at 68–72%, while on free-form benchmarks it disperses to 31–41%, with no overlap across all five benchmarks. Model×item interaction (up to 36%) is the second-largest component in cross-model analysis, consistent with prior findings on differential item functioning. Under binary scoring, BF16 and FP16 numerical precisions are functionally interchangeable (<0.1% interaction). Multi-facet designs require 10× fewer items than single-condition designs to reach a generalizability coefficient of 0.80 (55 versus 552 items), and cross-model projections confirm the ratio exceeds 10× across all benchmarks, with free-form benchmarks showing nearly double the efficiency gain of multiple-choice benchmarks. Within-family scale validation (3B–9B) confirms item dominance is scale-robust. We release an open-source G-theory pipeline and D-study budget calculator for evaluation design optimization.

1 Introduction

On multiple-choice benchmarks, item selection drives 34–38% of evaluation variance while prompt wording is negligible (<3%). On the free-form reasoning task GSM8K, the structure shifts: model×prompt interaction alone accounts for approximately 9%, while item selection drops to 17%. These numbers, drawn from our factorial analysis of 2.3 million evaluation records across five benchmarks, reveal that the variance structure of LLM evaluation is not a fixed property of the pipeline but shifts qualitatively with the response format. Current evaluation practice treats variance sources as independent nuisance factors and controls them one at a time: one study perturbs prompts, another varies precision, a third changes item subsets. This piecemeal approach cannot capture interactions between sources, nor can it predict how the relative importance of each source changes across evaluation settings. Leaderboard rankings built on single-condition evaluations are therefore sensitive to methodological choices that remain unreported and uncontrolled (Rodriguez et al., 2021; Dehghani et al., 2021). No unified factorial framework exists to decompose these sources simultaneously, quantify their interactions, and compare the resulting variance structure across benchmarks.

Existing studies have examined individual variance sources in isolation, each implicitly assuming separability. Sclar et al. (2024) documented up to 76% accuracy swings from prompt formatting perturbations but held hardware and item selection fixed. BrittleBench (Romanou et al., 2026) found 50% variance from prompt perturbation under the same constraint. Yuan et al. (2025) showed that numerical precision and GPU type cause up to 9% accuracy variation but provided no factorial attribution to specific sources or their interactions. Messing (2026) applied Generalizability theory (G-theory) to LLM evaluation and decomposed variance across five methodology facets on a single benchmark; judge identity dominated at 43.8%, but the study included no cross-benchmark comparison and excluded infrastructure facets such as numerical precision. Madaan et al. (2024) proposed per-factor variance metrics but did not cross factors. Studies of answer ordering (Pezeshkpour and Hruschka, 2024) and hyperparameter sensitiv-

ity (Dodge et al., 2020; Henderson et al., 2018) document additional sources in isolation. IRT approaches (Maia Polo et al., 2024; Lalor et al., 2019; Zhou et al., 2026) model item difficulty to construct efficient benchmarks but do not account for prompt, seed, or ordering facets. Bouthillier et al. (2021) established ANOVA-based variance decomposition for ML reproducibility, and G-theory has been applied to LLM-based assessment (Wei et al., 2025), but neither crosses infrastructure and methodology facets across multiple benchmarks. Three gaps remain: no study has (a) crossed infrastructure and methodology facets in a unified factorial design, (b) compared variance structures across response formats, or (c) translated variance components into design-dependent reproducibility budgets.

We address these gaps with a multi-benchmark G-theory study (Shavelson and Webb, 1991; Brennan, 2001) that crosses up to seven facets (item, model, numerical precision, prompt template, random seed, sampling temperature, and answer ordering) across eight open-weight 7–9B parameter models and five benchmarks (MMLU, ARC-Challenge, HellaSwag, GSM8K, and MATH), totaling 2.3 million records. Our results suggest that the variance structure of LLM evaluation is format-associated. We define *item-related variance* as the sum of item difficulty and model×item interaction (differential item functioning, DIF; §3.1). Across all three multiple-choice benchmarks, item-related variance concentrates at 68–72%, while on free-form benchmarks (GSM8K, MATH) it disperses to 31–41%, with no overlap across all five benchmarks. Model×item interaction (up to 36%) is the second-largest variance source in our multi-facet decomposition, consistent with prior IRT findings on differential item functioning (Pezeshkpour and Hruschka, 2024; Maia Polo et al., 2024). Models differ not in overall ability but in which specific items they find difficult. This format-associated pattern implies distinct optimization strategies: for MC benchmarks, item pool composition is the primary reliability lever; for free-form benchmarks, model identity and model×prompt interaction matter as much as item selection. D-study projections reveal that multi-facet designs require 10× fewer items than single-condition designs to achieve $G \geq 0.80$ (55 versus 552 items), with the efficiency gain itself being format-associated across benchmarks. Figure 2 visualizes the contrast across all five benchmarks.

Our contributions are as follows:

1. **Format-associated item-related variance concentration.** In a crossed 8-model × 5-benchmark design (2,304,000 records), item-related variance (item difficulty + model×item DIF) concentrates in MC benchmarks (68–72%) and disperses in FF benchmarks (31–41%), with no overlap across all five benchmarks. The mechanism is item difficulty heterogeneity: within MMLU's 57 subjects, difficulty dispersion correlates with item variance share ($\rho = 0.806$, $p = 0.005$).

2. **Design-dependent reproducibility budgets.** Multi-facet evaluation designs require 10.0× fewer items than single-condition designs to achieve $G \geq 0.80$ (55 versus 552 items). Cross-model D-study projections confirm the ratio exceeds 10× across all benchmarks and reveal a format-associated efficiency gap: FF benchmarks show ∼21× versus ∼12× for MC, indicating that single-condition evaluation of reasoning tasks incurs nearly double the reliability penalty.

3. **Precision separability (confirmatory).** Under binary scoring, BF16 and FP16 yield functionally identical results (<0.1% interaction with all other facets). This separability is metric-conditional: text-exact-match scoring reveals a 1.3% precision×temperature interaction absent under binary scoring.

4. **Practitioner toolkit.** We release a G-theory variance decomposition pipeline and D-study budget calculator. Practitioners can compute the number of items, models, and conditions needed for a target reliability before data collection.

2 Related Work

Variance Decomposition in ML/NLP Evaluation. Factorial decomposition of evaluation variance has roots in the ML reproducibility literature. Bouthillier et al. (2021) introduced an ANOVA-based template that decomposes benchmark variance across implementation details, hyperparameters, and data splits, establishing that design choices account for a substantial share of reproducibility failures. Several subsequent studies applied similar decomposition logic to specific facets of model evaluation (Dodge et al., 2020; Henderson et al., 2018; Magnusson et al., 2023). Messing (2026) brought Generalizability theory into

LLM evaluation, decomposing variance across five methodology facets on a single benchmark; judge identity dominated at 43.8%, with prompt×judge interaction at 12.4%. However, this study used a single benchmark, included no infrastructure facets such as numerical precision, and fixed the judge as an LLM evaluator rather than isolating other sources. Madaan et al. (2024) proposed per-factor variance metrics for LLM evaluation benchmarks but did not cross factors simultaneously. Recently, G-theory has been applied to LLM-based writing assessment (Wei et al., 2025) and annotation reliability (Anonymous, 2025), but these studies address scoring consistency rather than benchmark evaluation variance. Kunievsky and Evans (2025) proposed an alternative decomposition that separates intent uncertainty, articulation variance, and model uncertainty, but this framework does not cross infrastructure and methodology facets in a factorial design. Our work extends these approaches by crossing seven facets across five benchmarks, enabling direct comparison of variance structures across evaluation formats.

Infrastructure and Prompt Variance. A parallel line of work examines individual variance sources in isolation. Yuan et al. (2025) documented up to 9% accuracy variation from numerical precision; Messina and Scotta (2026) and He and Thinking Machines Lab (2025) formalized and mitigated CUDA nondeterminism from floating-point non-associativity. On the methodology side, Sclar et al. (2024) and BrittleBench (Romanou et al., 2026) found 50–76% accuracy swings from prompt formatting; Hua et al. (2025) showed that much of this sensitivity is an artifact of log-likelihood scoring (our generation-based protocol controls for this). Pezeshkpour and Hruschka (2024), Bui et al. (2025), and Biderman et al. (2024) documented ordering, seed, and training sensitivity respectively. Each study isolates one factor, precluding interaction estimation within a unified design.

IRT and Efficient Benchmarking. Item Response Theory (IRT) offers a complementary perspective on evaluation design by modeling item difficulty and discrimination parameters. Lalor et al. (2019) introduced IRT-based item analysis for NLP benchmarks, and Rodriguez et al. (2021) showed that IRT reveals hidden properties of evaluation items that raw accuracy scores obscure. Building on this foundation, Maia Polo et al. (2024) developed tinyBenchmarks, using IRT-based subsampling to reduce evaluation cost while preserving ranking accuracy, and Zhou et al. (2026) recommended a 116-item subset via PSN-IRT for efficient MMLU evaluation. Perlitz et al. (2023) and Kipnis et al. (2025) explored related strategies for benchmark compression; Li et al. (2025) extended IRT to adaptive testing. These approaches optimize item selection for latent trait estimation but do not model variance from prompt, seed, or ordering facets. IRT-selected items evaluated under a single methodological condition can therefore systematically underestimate the number of items required for reliable measurement across diverse evaluation settings.

Psychometrics Foundations. G-theory (Shavelson and Webb, 1991; Brennan, 2001) provides the canonical framework for multi-facet reliability analysis. Extensions to binary data include Mushquash and O'Connor (2006) and logistic-link generalized G-theory (Jiang and Skorupski, 2018; Tuerlinckx et al., 2006). We operationalize differential item functioning (Embretson and Reise, 2000) as model×item interaction. Recent psychometric perspectives on LLM evaluation (Xiao et al., 2025; Jin et al., 2025) address complementary concerns of construct validity and measurement properties.

Positioning. Relative to Messing (2026), we fix the judge (generation-based scoring) to isolate the remaining facets and extend to multiple benchmarks, revealing format-associated structure. Relative to IRT approaches (Maia Polo et al., 2024; Zhou et al., 2026), G-theory optimizes an orthogonal dimension: conditions needed for target reliability rather than item discrimination. Relative to isolated-factor studies (Yuan et al., 2025; Sclar et al., 2024; Romanou et al., 2026; Madaan et al., 2024), our factorial crossing enables interaction estimation.

3 Method

3.1 Generalizability Theory

Generalizability theory (G-theory) treats each measurement as one observation from a universe of admissible conditions defined by the crossed facets of an evaluation design (Shavelson and Webb, 1991; Brennan, 2001). The *object of measurement* is the entity whose consistency is of interest. In our setting, test items (p) serve as the object of measurement, and the goal is to determine how reliably

item-level scores generalize across the methodological conditions under which they are collected.

A *G-study* decomposes total observed variance into components σ_x^2 for each facet x and their interactions. Each component quantifies how much variability that source contributes to the overall measurement. Large interaction components (e.g., σ_{pM}^2 for item-by-model) indicate that facets do not affect items uniformly but instead create differential difficulty profiles. We estimate variance components via Henderson Method I, an unbiased ANOVA-based estimator for balanced designs (Brennan, 2001, Ch. 3), with 1,000-resample bootstrap confidence intervals (item-level resampling with replacement, preserving the fully-crossed structure across all other facets) (Li, 2023).

A *D-study* projects the generalizability coefficient under hypothetical designs with different numbers of conditions per facet:

$$G_{\text{item}} = \frac{\sigma_p^2}{\sigma_p^2 + \sigma_{\text{rel}}^2} \tag{1}$$

where σ_{rel}^2 aggregates all item-by-facet interaction variances, each divided by the number of conditions for the corresponding facets (Brennan, 2001, Ch. 4). This prospective capability distinguishes G-theory from classical ANOVA: rather than only attributing variance post hoc, a D-study enables optimization of evaluation designs *before* data collection (Shavelson et al., 1989). We define *item-related variance* as the sum of the item main effect (σ_p^2) and the model×item interaction (σ_{pM}^2). The latter captures differential item functioning (DIF): items whose difficulty ranking changes across models (Brennan, 2001, Ch. 4). Because DIF is operationally an item-level property—it reflects model-specific item difficulty profiles rather than a global model ability shift—aggregating $\sigma_p^2 + \sigma_{pM}^2$ provides a unified measure of item-driven variance.

In ML terms, a G-study is a multi-way ANOVA where each facet is a design variable and variance components quantify each variable's contribution; a D-study projects the number of conditions per facet needed for a target reliability G, analogous to a signal-to-noise ratio where signal is item difficulty variance and noise is everything that changes across evaluation conditions.

Throughout this paper, we use the following notation: p = item, M = model, π = numerical precision, δ = prompt template, s = random seed, τ = sampling temperature, o = answer ordering, and σ_x^2 = variance component for source x.

3.2 Experimental Design: MMLU

We design a fully crossed factorial experiment (exp-001) that simultaneously varies seven facets of LLM evaluation on MMLU. Table 1 summarizes the complete design matrix, and Figure 1 illustrates the crossing structure.

Facets. Items (p): 200 MMLU questions selected via purposive stratified sampling across 10 of MMLU's 57 subjects (20 per subject, deterministic selection with seed 42). The 10 subjects span four disciplinary domains—STEM: abstract algebra, college physics, high school mathematics, computer security; humanities: world religions, philosophy; social sciences: US foreign policy, international law; professional: professional medicine, clinical knowledge—ensuring that variance estimates are not dominated by a single knowledge domain. Models (M): eight open-weight 7–9B parameter instruction-tuned models spanning eight architecture families—DeepSeek-7B, Gemma-2-9B, InternLM2.5-7B, Llama-3.1-8B-Instruct, Mistral-7B-v0.3-Instruct, OLMo-2-7B, Qwen3-8B, and Yi-1.5-9B (full identifiers in Appendix C)—treated as a quasi-random facet (§3.6). Numerical precision (π): BF16, FP16, and FP32, all evaluated with `enforce_eager=True` for fair comparison across precision levels. Prompt template (δ): six surface perturbations of the evaluation instruction (Sclar et al., 2024), varying option layout (lettered vs. numbered), instruction wording (direct vs. contextual), and answer prompt cue for MC benchmarks; for FF benchmarks, templates vary chain-of-thought elicitation strategy (full template text in Appendix C). Random seed (s): six fixed values (42, 123, 456, 789, 1024, 2048). Sampling temperature (τ): three levels (0.0, 0.3, 0.7) spanning deterministic to moderately stochastic decoding. Answer ordering (o): four permutations of few-shot demonstration examples (for MMLU; structurally inert for other benchmarks, see §3.3) (Pezeshkpour and Hruschka, 2024).

Sample sizes. The cross-model BF16 balanced subset yields 460,800 records (8 models × 200 items × 6 prompts × 6 seeds × 2 temperatures × 4 orderings). A single-model analysis (Llama) spans all three precisions, yielding 259,200 binary observations across 1,296 fully balanced conditions. The BF16 subset serves as the primary cross-model sample because BF16 is the standard inference precision for these architectures.

Table 1: Fully crossed G-theory design. Exp-001 evaluates MMLU across all facet levels; exp-002 extends to five benchmarks with reduced crossing. †BF16 balanced subset for cross-model analysis.

Facet	Levels	exp-001	exp-002
Item (p)	Stratified sample	200	200×5
Model (M)	8 architecture families	8	8
Precision (π)	BF16 / FP16 / FP32	3	1
Prompt (δ)	Paraphrases	6	6
Seed (s)	Fixed set	6	6
Temperature (τ)	0.0 / 0.3 / 0.7	3	2
Ordering (o)	Demo permutations	4	1
Total records		460,800†	2,304,800

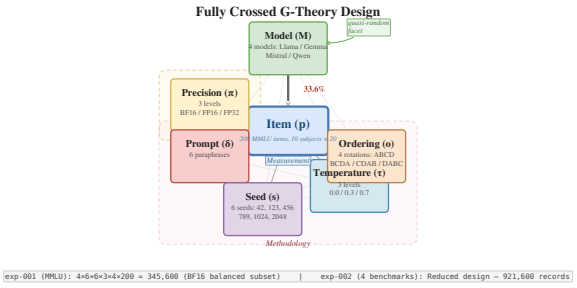


Figure 1: Fully crossed G-theory design for LLM evaluation. Each observation is a binary correctness score determined by the crossing of seven facets: item (p), model (M), precision (π), prompt (δ), seed (s), temperature (τ), and ordering (o).

3.3 Cross-Benchmark Extension

To test whether the variance structure observed on MMLU generalizes across evaluation formats, exp-002 extends the design to five benchmarks: MMLU, ARC-Challenge, and HellaSwag (multiple-choice), and GSM8K and MATH (free-form generation). The design crosses 8 models × BF16 × 2 temperatures (0.0, 0.7) × 6 prompts × 6 seeds × 4 orderings × 200 items per benchmark, totaling 2,304,000 records. This reduced crossing fixes precision at BF16 and uses two temperature levels to concentrate statistical power on format-associated differences. The ordering facet operates only on MMLU, where it selects among four permutations of the few-shot demonstration examples. For all other benchmarks, the prompt construction does not use the ordering parameter; ordering is structurally inert ($\sigma^2_{\text{ordering}} = 0$ by construction) and is retained solely for design balance. The two temperature levels bracket the most common evaluation configurations: deterministic ($\tau = 0.0$) and a moderate stochastic setting ($\tau = 0.7$) representative of practical deployment. A subset analysis confirms that this reduced crossing preserves the variance structure of the full design ($\Delta G = 0.0008$; Appendix A).

3.4 Evaluation Protocol

All evaluations use generation-based scoring (free-text response with answer extraction) rather than log-likelihood ranking, following Hua et al. (2025). Binary correctness (1/0) is the primary metric; text-exact-match is secondary. Inference uses vLLM 0.9.2 without batch-invariant kernels, preserving CUDA-level nondeterminism as a measurable variance source (Messina and Scotta, 2026; He and Thinking Machines Lab, 2025). Few-shot protocols: 5-shot MMLU/ARC, 4-shot MATH, 8-shot GSM8K, 10-shot HellaSwag.

3.5 Binary Data Assumptions

Standard G-theory assumes continuous, approximately normal outcome data (Shavelson and Webb, 1991). Our binary correctness scores (0/1) violate this assumption. We establish robustness through simulation and estimator properties, with two additional analyses (GLMM complement, accuracy-range stability) in Appendix A.

A probit simulation matched to our 1,296-condition design confirms that Henderson Method I recovers variance component rankings from dichotomized data (Spearman $\rho = 0.732$ between binary and continuous rankings; `item_id` dominant in all 100 replications). Henderson Method I produces unbiased estimates for balanced designs regardless of outcome distribution (Brennan, 2001, Ch. 12), and REML estimates differ by <0.5pp on all components (Appendix A).

Treating eight purposively selected models as a quasi-random facet is validated by two analyses (Appendix A): four-model subset D-study projections predict $G = 0.893$ at eight models (observed: $G = 0.896$, $\Delta = 0.003$), and all eight leave-one-model-out subsets preserve component rankings.

4 Experiments

All experiments use open-weight 7–9B parameter models evaluated with vLLM 0.9.2 on a single NVIDIA RTX PRO 6000 GPU (96GB) with binary correctness scoring.

4.1 Single-Model Baseline

We first establish the variance structure within a single model (Llama-3.1-8B-Instruct) on MMLU

to isolate methodological facet contributions before introducing cross-model variation.

Design. The G-study crosses six facets: 200 items × 3 numerical precisions (BF16, FP16, FP32) × 6 prompt templates × 6 random seeds × 3 temperature levels (0.0, 0.3, 0.7) × 4 answer orderings, yielding 259,200 binary observations. Variance components are estimated via Henderson Method I with bootstrap confidence intervals (1,000 resamples).

Results. Under binary scoring, item difficulty (`item_id`) accounts for 58.4% of total variance, with the residual term contributing 23.1%. All main facet effects are negligible (<1% each): precision 0.9%, prompt template, temperature, seed, and ordering each <0.5%. The interesting structure appears in item-conditional interactions: prompt×item is the largest at 5.5%, followed by seed×item (4.7%), precision×item (3.3%), temperature×item (2.9%), and ordering×item (0.8%). This pattern indicates that methodological choices do not shift performance uniformly across items; instead, specific items are differentially sensitive to specific facets—prompt template variation produces the largest item-conditional effect.

The resulting generalizability coefficient is $G_{\text{item}} = 0.936$, indicating that 200 items under multi-facet conditions produce highly reliable rankings. A D-study projection shows $G \geq 0.80$ is achievable with as few as 55 items in this multi-facet design.

For text-exact-match scoring (requiring verbatim string matches), the variance structure shifts dramatically: temperature dominates at 79.7% of variance, and G_{item} collapses to 0.007. This metric-conditional divergence motivates our exclusive use of binary correctness for the remaining analyses.

4.2 Cross-Model Variance

Extending to eight models (BF16 subset, 460,800 records), Table 2 shows that item difficulty drops from 58.4% to 37.9%, while model×item interaction emerges at up to 36%. The model main effect is 3–5% (MC) and 9–11% (FF), confirming that models differ primarily in *which items* they find difficult rather than in overall ability level.

The per-model interaction magnitudes reveal meaningful asymmetry, suggesting that instruction-tuning strategies produce model-specific item difficulty profiles.

Table 2: Variance decomposition for the cross-model MMLU design (8 models, BF16, 460,800 records). Components estimated via Henderson Method I. $G_{\text{item}} = 0.896$.

Component	Var.%
item_id	**37.9**
model×item	30.2
residual	25.0
model	2.8
prompt×item	2.5
model×prompt	1.6
all others	<1.0

4.3 Format-Associated Variance Structure

We extend to five benchmarks (three MC: MMLU, ARC-Challenge, HellaSwag; two FF: GSM8K, MATH) using exp-002 (8 models × BF16 × 2 temperatures × 6 prompts × 6 seeds × 4 orderings × 200 items; MATH uses 7 models due to incomplete DeepSeek inference data). Table 3 reveals a systematic format-associated pattern. For MMLU and ARC-Challenge, item difficulty and model×item are the two largest components (37.9% + 30.2% and 37.9% + 33.7% respectively). HellaSwag is the sole MC exception where model×item slightly exceeds item (36.0% vs. 34.3%), though their sum remains in the MC range. In all three MC benchmarks, item-related variance (item + model×item DIF; e.g., MMLU: 37.9% + 30.2% = 68.1%) concentrates at 68–72%.

Free-form benchmarks differ markedly. On GSM8K, variance distributes across item (16.9%), model×item (14.6%), model (10.7%), and model×prompt (8.5%); prompt main effect is 5.4%. MATH shows an intermediate pattern: item leads at 24.0% with model×item at 17.3%, but residual variance is substantially larger (43.9%). The large FF residuals (40–44%) likely reflect chain-of-thought path diversity, answer extraction variability, and unmodeled higher-order interactions.

The observed variance structure is format-associated: MC benchmarks concentrate item-related variance at 68–72%, while FF benchmarks disperse it to 31–41%, with no overlap. Unbundled, `item_id` alone separates MC (34–38%) from FF (17–24%) with no overlap. A permutation test on `item_id` variance shares yields $p = 0.10$ ($n = 5$, $\binom{5}{3} = 10$ exhaustive permutations). The item-level effect size is tightly bounded (Cohen's $d = 1.73$; bootstrap 95% CI $[1.72, 1.83]$), but the benchmark-level CI spans zero ($[-1.7, 5.1]$, non-central t, $df = 3$), reflecting the inherent limita-

Table 3: Format-associated variance structure across five benchmarks (exp-002, 8 models; MATH: 7). Variance components (%) per benchmark; MC = multiple-choice, FF = free-form. Bold indicates the dominant non-residual component per benchmark. For MC benchmarks, "prompt" reports prompt×item (prompt main effect <1%); for FF benchmarks, it reports the prompt main effect.

	MC			FF	
	MMLU	ARC	Hella.	GSM	MATH
item	**37.9**	**37.9**	34.3	16.9	24.0
mod.×item	30.2	33.7	**36.0**	14.6	17.3
model	2.8	3.1	5.4	10.7	8.5
prompt	2.5	0.4	2.3	5.4	1.0
mod.×prmpt	1.6	8.0	0.6	8.5	2.0
residual	25.0	15.4	21.4	39.9	43.9
all others	<1.0	1.5	<1.0	4.0	3.3

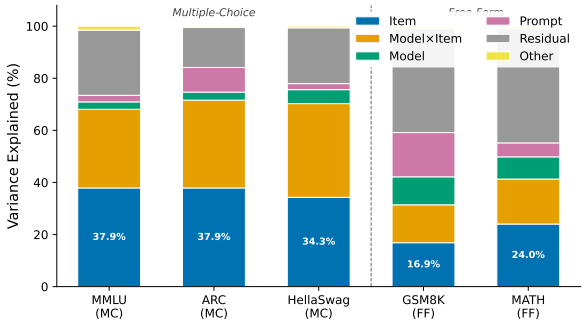


Figure 2: Variance decomposition (Henderson Method I) across five benchmarks and eight models (MATH: seven). Multiple-choice benchmarks share an item-dominated structure; HellaSwag is the exception where model×item (36.0%) exceeds item difficulty (34.3%). Free-form benchmarks (GSM8K, MATH) distribute variance across item, model, and interaction components with no single dominant source.

tion of $n = 5$. The contrast between item-level precision and benchmark-level uncertainty underscores that the format-associated pattern is reliably measured within our five benchmarks but requires additional benchmarks for confirmatory generalization. This structural difference does not extend to item budgets: GSM8K requires 128 items for $G \geq 0.80$—the highest of all benchmarks—while HellaSwag (MC, 118) exceeds both ARC (MC, 98) and MATH (FF, 101), demonstrating that format alone does not predict evaluation cost. Figure 2 visualizes the contrast.

To probe the mechanism behind item variance, we exploit the natural heterogeneity within MMLU's 57 subjects. Subjects with higher item difficulty dispersion (measured by std(p_i) across models) exhibit higher item variance shares, with Spearman $\rho = 0.806$ ($p = 0.005$, $n = 10$ subjects). Notably, `high_school_mathematics`, the only MC subject with open-ended numerical items, shows item variance of just 10.4%—below the free-form benchmark GSM8K (16.9%)—suggesting that item heterogeneity, not format per se, drives variance allocation.

4.4 Precision Separability

Under binary scoring, BF16 and FP16 are functionally interchangeable: mean accuracy differs by 0.23 percentage points, and all precision×methodology interactions contribute <0.1% of variance (Yuan et al., 2025). Under text-exact-match scoring, a precision×temperature interaction emerges at 1.3%, consistent with the metric-conditional pattern in §4.1. For standard binary evaluation at BF16/FP16 on single-GPU inference, numerical precision is a negligible variance source that need not be controlled (Shanmugavelu et al., 2024).

4.5 Design-Dependent Reproducibility Budgets

The previous sections establish that item selection and model×item interaction dominate evaluation variance. A natural question follows: how many items does a given evaluation design need to achieve a target reliability? We address this through D-study projections that translate variance components into concrete item budgets.

Table 4 compares three evaluation designs at the $G \geq 0.80$ threshold using the single-model (Llama-3.1-8B-Instruct) MMLU variance components. The multi-facet design (1,296 conditions per item: 3 precisions × 6 prompts × 6 seeds × 3 temperatures × 4 orderings) achieves $G = 0.936$ at 200 items and requires only 55 items for $G \geq 0.80$. A single-condition design (one fixed prompt, seed, ordering, and precision per item) reaches only $G = 0.592$ at 200 items and requires 552 items for the same threshold. The ratio between these budgets is 10.0×, and this ratio remains stable across subset sizes from 50 to 200 items (CV = 1.9%).

An illustrative comparison: Zhou et al. (2026)'s IRT-selected 116 items yield $G = 0.457$ under single-condition protocols but $G = 0.895$ under multi-facet conditions. IRT item selection and G-theory protocol optimization are orthogonal; combining both achieves reliable measurement with fewer items. The higher total evaluation count of multi-facet designs (71,280 vs. 552) reflects the cost of reliable measurement; single-condition

Table 4: D-study reproducibility budgets under different evaluation designs (single-model, MMLU). Total evals = items × conditions.

Design	Cond.	$n_{G\geq.80}$	Total	G_{200}	G_{116}
Multi-facet	1,296	**55**	71,280	**.936**	—
Single-cond.	1	552	552	.592	.457
IRT+multi	1,296	—	150,336	—	.895‡
IRT baseline	—	116	116	—	.457†

† PSN-IRT 116 items (Zhou et al., 2026) under single-condition: $G = 0.457$.
‡ Same 116 items under multi-facet: $G = 0.895$.

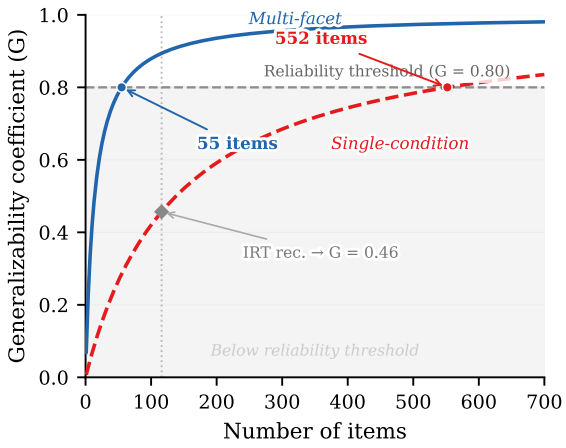


Figure 3: D-study projection curves (single-model, MMLU). The multi-facet design (10.0× fewer items) reaches $G \geq 0.80$ at 55 items; the single-condition design requires 552 items. Cross-model per-benchmark projections are in Appendix B. Shaded band: 95% bootstrap CIs.

$G = 0.592$ falls below acceptable thresholds.

Cross-model D-study projections (Appendix B) reveal that the efficiency gap is itself format-associated. MC benchmarks exhibit multi-facet/single-condition ratios of 11–13×, while FF benchmarks show 21–22× (bootstrap CIs confirm all ratios exceed 10×). Single-condition evaluation of free-form benchmarks incurs nearly double the reliability penalty of multiple-choice benchmarks, making multi-facet designs even more critical for reasoning tasks where variance distributes across model, prompt, and interaction components (§4.3).

Figure 3 plots the D-study projection curves for each design. The multi-facet curve saturates rapidly (diminishing returns beyond ∼150 items), while the single-condition curve rises slowly and does not reach $G = 0.80$ within typical benchmark sizes.

4.6 Preliminary Scale Validation

To test whether item dominance is scale-robust, we conduct a within-family analysis using Qwen2.5 at two parameter scales (3B-Instruct and 7B-Instruct) on MMLU under the same BF16 × 2-temperature × 6-prompt × 6-seed × 4-ordering design. Using a single model family isolates scale as the sole variable. At both 3B (accuracy 0.682) and 7B (0.742), `item_id` is the dominant variance component: 75.3% and 73.4%, respectively (cf. Llama-3.1-8B single-model: 58.4–68.5% depending on facet crossing). Item dominance is preserved across scales, with minor scale-dependent prompt sensitivity (prompt×item: 12.2% at 7B vs. 8.8% at 3B). Generalizability coefficients ($G = 0.963$ and 0.956) confirm that 200 items yield reliable measurements at both scales. Future work will extend this validation to 70B+ models via a fully crossed multi-scale G-study.

4.7 Robustness Checks

Three independent 200-item samples from the full MMLU pool yield item difficulty variance shares of 73.2%, 72.6%, and 69.0% (CV = 3.1%) and G_{item} values with CV = 0.5%, confirming sampling stability. Seven additional analyses (GLMM complement, accuracy-range stability, estimator comparison, CI method, facet specification, leave-one-model-out, and null calibration) are reported in Appendix A; all confirm the stability of the reported variance structure.

5 Conclusion

Our results suggest that the variance structure of LLM evaluation is format-associated: item-related variance (item + model×item DIF) concentrates at 68–72% on MC benchmarks and disperses to 31–41% on FF benchmarks, with no overlap. For MC benchmarks, item pool composition is the primary reliability lever; for FF benchmarks, model identity and model×prompt interaction matter equally. Multi-facet designs require 10× fewer items than single-condition designs ($G \geq 0.80$: 55 vs. 552 items), with FF benchmarks showing nearly double the efficiency gain (∼21× vs. ∼12×). BF16 and FP16 are interchangeable under binary scoring. The released D-study calculator enables practitioners to determine reproducibility budgets before data collection. Code, data, and the D-study budget calculator will be released upon publication.

6 Limitations

Our findings are bounded by three scope constraints. First, the primary cross-model anal-

ysis covers eight open-weight 7–9B parameter instruction-tuned models; within-family scale validation extends to 3B–9B (Qwen2.5), but results do not generalize to API-based models (where infrastructure is opaque), base (non-instruction-tuned) models, models above 9B parameters, or domain-adapted fine-tuned variants. Extending this validation to larger scales (e.g., 70B+) remains for future work; larger models with higher accuracy could exhibit compressed item variance due to ceiling effects. Second, all experiments run on single-GPU inference (NVIDIA RTX PRO 6000); multi-GPU tensor parallelism introduces additional numerical non-associativity that our design does not capture. Third, the benchmark sample covers three multiple-choice and two free-form tasks ($n = 5$); item-level variance patterns within each benchmark are well-powered, but the format-associated claim rests on a benchmark-level comparison with limited statistical power ($p = 0.10$, permutation test). Other evaluation formats such as code generation, long-form text production, and multi-turn dialogue may exhibit different variance structures. Two natural extensions remain: multi-GPU tensor parallelism (testing precision separability under distributed numerical non-associativity) and fine-tuning variance analysis (quantifying how adaptation reshapes item difficulty profiles).

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A Additional Robustness Analyses

We report eight robustness analyses that verify the stability of the variance decomposition under alternative analytical choices.

GLMM complement. A generalized linear mixed model (GLMM) with marginal logit link produces variance component rankings that perfectly preserve all seven grouped component orderings ($\rho = 1.0$) relative to Henderson Method I (Jiang and Skorupski, 2018; Tuerlinckx et al., 2006). This confirms that the linear variance decomposition is robust to the binary outcome distribution.

Accuracy-range stability. The `item_id` variance share remains stable across subsets with mean accuracy from 0.62 to 0.72 ($r = -0.048$). This rules out distortion from binary floor or ceiling effects at observed accuracy levels.

Estimator robustness. We compare Henderson Method I with restricted maximum likelihood (REML-EM, 500 iterations, practical convergence $\Delta < 0.003$ percentage points). All variance components differ by <0.5 percentage points between estimators. The largest discrepancy occurs on the residual component (0.47pp); all substantive components (item, model×item, model) differ by <0.3pp. Henderson Method I is preferred for our balanced designs because it produces unbiased estimates without distributional assumptions (Brennan, 2001, Ch. 12).

CI method comparison. Bootstrap (1,000 item-level resamples with replacement) and delete-1 jackknife confidence intervals agree closely: width ratios range from 0.96 to 1.04 across all variance components. The agreement indicates that the dependence structure introduced by item-level bootstrap resampling (which creates ∼127 unique items per 200-item resample) does not materially distort interval estimates.

Facet specification sensitivity. Treating temperature and ordering as fixed (rather than random) facets changes G_{item} by only 0.37 percentage points (all-random: 0.820 vs. fixed temperature+ordering: 0.824). The small difference reflects the negligible temperature×item (0.65%) and ordering×item (0.54%) variance shares. The random-facet assumption is therefore robust to facet classification.

Leave-one-model-out (LOMO). All eight seven-model subsets preserve the item + model×item dominance pattern: item_id ranges from 36% to 41%, and G_{item} ranges from 0.877 to 0.889. No single model disproportionately drives the observed variance structure. The consistency of component rankings across all subsets supports the quasi-random model facet assumption (§3.5).

Null calibration. We generate 100 Bernoulli-distributed datasets with item difficulties matched to the empirical distribution but no systematic facet effects. The observed-to-null variance ratio is 1.0008, confirming that Henderson Method I does not introduce systematic bias when applied to binary outcomes. The null simulation also verifies that the item variance share (58.4%) is not an artifact of the Bernoulli $p(1-p)$ ceiling: the null produces item variance of 0.08%, three orders of magnitude below the observed value. Across all 100 replications, the same three components (item, model×item, residual) emerge as the top three (set consistency 100%), with Kendall's $\tau = 0.81$ between binary and continuous rankings.

Reduced crossing validation. To verify that the reduced crossing in exp-002 (BF16 only × 2 temperatures) does not introduce confounding, we extracted the corresponding subset from the fully crossed exp-001 design. The generalizability coefficient differs by only 0.0008 ($G = 0.9361$ vs. 0.9369), and all non-precision variance components differ by less than 2 percentage points. The 10pp increase in item variance share (58.4% → 68.5%) is fully explained by the removal of precision-related components (∼4.5% total), which redistributes across remaining sources. This confirms that the exp-002 design preserves the variance structure of the full crossing.

B Cross-Model D-Study Projections

Table 5 extends the single-model D-study (Table 4 in main text) to per-benchmark cross-model projections with bootstrap confidence intervals (1,000 item-level resamples, 8 models, 2,304 conditions per item). The efficiency ratio is format-associated: MC benchmarks show 11–13× (mean 12.0× [11.6, 12.5]), while FF benchmarks show 21–22× (mean 21.5× [20.4, 22.8]).

Table 5: Cross-model D-study reproducibility budgets with 95% bootstrap CIs. $n_{G\geq 0.80}$: items needed. Multi = multi-facet (2,304 cond.), Single = single-condition.

	Multi $n_{G\geq 0.80}$		Single $n_{G\geq 0.80}$		Ratio
	Point	95% CI	Point	95% CI	[95% CI]
Multiple-choice					
MMLU	93	[76, 116]	1216	[1004, 1491]	13.1 [12.5, 13.7]
ARC	98	[71, 139]	1078	[793, 1512]	11.0 [10.7, 11.4]
Hella.	118	[93, 151]	1403	[1120, 1790]	11.9 [11.5, 12.3]
Free-form					
GSM8K	128	[103, 162]	2768	[2230, 3531]	21.6 [20.7, 23.0]
MATH	101	[86, 122]	2160	[1853, 2631]	21.4 [20.2, 22.5]

The higher item requirements relative to the single-model analysis (55 items in Table 4) reflect the additional model×item interaction variance that enters σ^2_{rel} when models are crossed as a facet. The format-associated ratio pattern—FF benchmarks showing ∼2× the MC ratio—is consistent with the distributed variance structure observed in §4.3, where FF benchmarks allocate more variance to model and model×prompt components. MATH results use verified per-model inference across 7 models; DeepSeek was excluded due to incomplete inference data (37,400/57,600 expected records).

C Model and Prompt Details

Models. Table 6 lists the eight models with their HuggingFace identifiers.

Table 6: Models used in all experiments.

Short name	HuggingFace identifier	Params
Llama-3.1-8B	`meta-llama/Llama-3.1-8B-Instruct`	8B
Gemma-2-9B	`google/gemma-2-9b-it`	9B
Mistral-7B-v0.3	`mistralai/Mistral-7B-Instruct-v0.3`	7B
Qwen3-8B	`Qwen/Qwen3-8B`	8B
Yi-1.5-9B	`01-ai/Yi-1.5-9B-Chat`	9B
OLMo-2-7B	`allenai/OLMo-2-1124-7B-Instruct`	7B
InternLM2.5-7B	`internlm/internlm2_5-7b-chat`	7B
DeepSeek-7B	`deepseek-ai/deepseek-llm-7b-chat`	7B

Prompt templates. Our six prompt templates follow a surface perturbation design inspired by BrittleBench (Romanou et al., 2026). For multiple-choice benchmarks, templates vary along three dimensions: option layout (lettered vs. numbered), instruction wording (direct vs. contextual), and answer prompt cue. For free-form benchmarks, templates vary chain-of-thought elicitation strategy. All six templates are semantically equivalent—they request the same task with different surface forms—to isolate formatting effects from content effects. ARC-Challenge and HellaSwag use the same six surface patterns as MMLU with domain-appropriate system instructions; MATH mirrors the GSM8K pattern with `\boxed{}` answer formatting.

MMLU templates (representative MC format):

1. `The following are multiple choice questions (with answers) about {subject}.\n\n{question}\nA. ... D. ...\nAnswer:`
2. `Answer the following {subject} questions by selecting the correct option.\nQuestion: {question}\n(A) ... (B) ... (C) ... (D) ...\nCorrect answer:`
3. `Below are {subject} multiple-choice problems. Pick the right letter.\nQ: {question}\n A) ... B) ... C) ... D) ...\nA:`
4. `Answer the following {subject} question by selecting A, B, C, or D.\n{question}\n- A. ... - B. ... - C. ... - D. ...\nAnswer:`
5. `Below is a {subject} exam question with four possible answers. Choose the correct one.\n{question}\nOption A: ... Option B: ... Option C: ... Option D: ...\nCorrect:`
6. `Question ({subject}):\n{question}\nChoice (A) ... (B) ... (C) ... (D) ...\nYour answer:`

GSM8K templates (representative FF format):

1. `Solve the following math problem step by step.\nQ: {question}\nA: Let's think step by step.`
2. `Answer the following math word problem. Show your work.\nQuestion: {question}\nSolution:`
3. `Solve each problem below. Write your reasoning, then give the final answer after ####.\nProblem: {question}\nReasoning:`
4. `You are a math tutor. Solve the problem and show your steps.\n{question}\nSteps:`
5. `Work through this math question carefully.\nQ: {question}\nWork:`
6. `Math problem:\n{question}\nLet's solve this step by step.`